Delivery Fleet Cost Modelling & Optimisation Report.

1. Introduction and Scenario Overview.

The optimal resource distribution has become one of the most important operational issues faced by the companies that deliver or provide logistics, especially in case there is no guarantee of the daily demand, and the punishment of the underperformance may be high. In this paper, a small delivery company with one depot will be taken into account and the number of vans to be deployed and the number of employees (drivers and loaders) to be scheduled every day will be considered. Such decisions have a direct impact on the overall daily operation cost and quality of the service offered by the company

The company is exposed to various types of costs. First are the fixed costs per van which are leasing, insurance, and depreciation. Second, there are labour costs on the scheduling of the employees, irrespective of the levels of demand per day (Gössling, Kees and Litman, 2022). Third, the cost of running (fuel and maintenance) is more expensive with the quantity of deliveries made. Lastly, when the required capacity is not given against demand, the company will face penalty expenses such as overtime payment, compensation of customers, refunds, and damage to reputation through delayed or missed deliveries.

The main aim of this analysis is three-fold. The mathematical and statistical model is first created to describe the dependence of daily total cost on the demand and staffing decisions. Second, the cost-minimising solution, involving both vans and employees is determined using the fitted model at a given level of demand. Third, the ambiguity of the projected cost at this optimal level of staffing is computed by means of bootstrap resampling. The analysis has been conducted with the help of Python and the given dataset.

2. Data Description

The data set has the following data, 500 observations per day with one row representing a single operating day. The following variables are registered on a daily basis:

demand: how many delivery jobs have been demanded.

vans: the amount of vans employed during that day.

employees: the number of drivers and loaders who will be employed.

total cost: the total operating cost per day (£) which consists of fixed, variable and penalty costs.

The main variables are summarised in Table 1 to show the important descriptive statistics.

Table 1: Summary Statistics

Variable	Mean	Std Dev	Min	Max
Demand	96.22	9.57	65	127
Vans	5.56	2.97	1	12
Employees	8.74	3.72	2	16
Total Cost (£)	2240.12	718.21	892.81	7048.71

The data indicate that the demand changes rather significantly and varies between 65 to 127 jobs per day. There is also a significant difference in the total cost, implying that the change in the demand and staff choices play a significant role in the operational cost.

3. Task 1 Understanding the Problem and Exploratory Analysis

Task 2 Research Project Planning and Introduction.

3.1 Problem Statement

The optimisation problem of the company can be summarised as follows:

Purpose: Maximise the anticipated total cost per day.

- Decision variables:
- Number of vans
- Number of employees

Uncertain factors:

- Daily demand for deliveries
- Day-to-day variation in costs, which is caused by the cost of fuel, congestion, overtime, and penalties.

It is difficult to strike a balance between fixed staffing costs and potentially high costs of penalties when the capacity is not adequate.

3.2 Correlation Analysis

	demand	vans	employees	total_cost
demand	1.000000	0.171627	0.214491	-0.158875
vans	0.171627	1.000000	0.048960	-0.539070
employees	0.214491	0.048960	1.000000	-0.331741
total_cost	-0.158875	-0.539070	-0.331741	1.000000

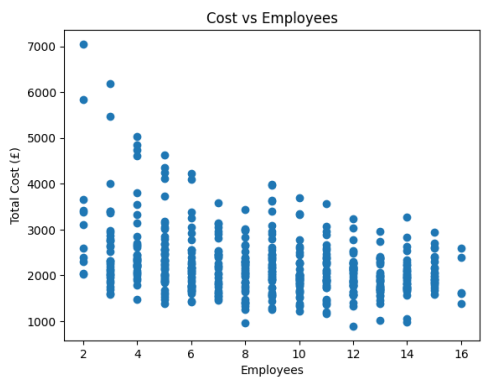
Simple Pearson correlations were calculated with demand, vans, employees, and total cost. The correlation of the total cost with vans (-0.54) and employees (-0.33) is negative, whereas the correlation between the total cost and demand is relatively low (-0.16).

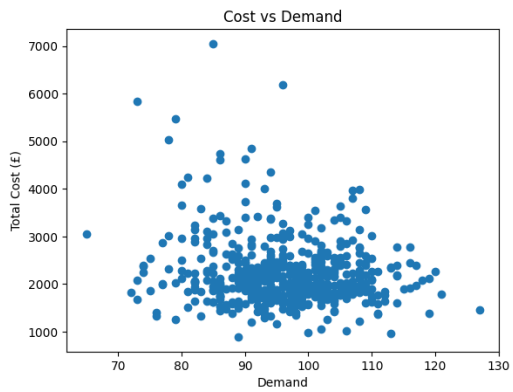
Such an outcome might look paradoxical at first sight, since the costs would supposedly be higher when more vans and workers are deployed. The negative correlations, however, indicate that the penalty costs are dominant: days where there are inadequate staffing compared to the demand experience very

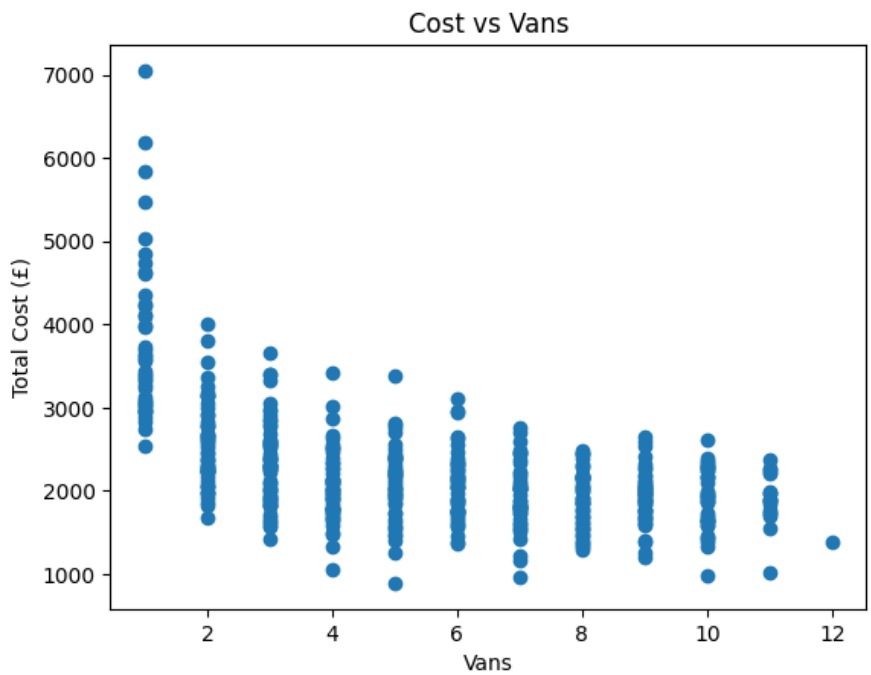
high overall costs, whereas days where there is an increase in staffing do not experience the penalties and hence experience lower overall costs (Senthilnathan, 2019).

3.3 Visual Exploration

The further exploration of the relationship between variables was conducted with the help of scatterplots. These visualisations indicate some interesting trends:







- A high demand level of a product has sharp increments in the total cost especially when personnel is minimal.
- When the number of vans and employees is increased, the extreme cost outcomes will become less frequent.
- It is apparent that the cost and staffing relationship is nonlinear and increases disproportionally to capital capacity being bound.

- These observations inspire the consideration of nonlinear aspects especially those associated with the use of the system in the cost model.

4. Task 2 -Cost Model Development/Selection.

4.1 Utilisation Concept

In order to measure congestion and capacity pressure, the definition of system utilisation is:

u=demand/(vans×employees)

This amount is the demand compared to the capacity. In the event of high utilisation, the system is working under or over capacity and results in overtime, delay, and penalty fees. It is thus logical to anticipate that utilisation of total cost will rise drastically.

4.2 Candidate Models

Three candidate regression models were suggested and estimated with the help of ordinary least squares (OLS).

Model 1: Linear Baseline

$C=\beta 0+\beta 1D+\beta 2V+\beta 3E$

This model only describes linear impacts of staffing and demand.

Model 2: Linear + Utilisation

$C=\beta 0+\beta 1D+\beta 2V+\beta 3E+\beta 4u$

Utilisation is another predictor that is introduced under this specification.

Model 3: Nonlinear Utilisation (Final Model)

$C=\beta 0+\beta 1D+\beta 2V+\beta 3E+\beta 4u+\beta 5u^2$

Squared utilisation term enables costs to grow nonlinearly with the capacity constraints when they are severe.

4.3 Model Comparison

The sum of squared errors (SSE) was used in comparing the models.

Model	Predictors	SSE
Model 1	Linear only	158,538,918
Model 2	+ utilisation	57,405,776
Model 3	+ utilisation²	54,538,811

The inclusion of the utilisation contributes greatly to the model fit and the error of the nonlinear model is the least.

4.4 Final Model Selection

Model 3 was chosen as the optimal model because it offers an optimal tradeoff between the model complexity and the power to explain. It records the steep rise of costs at large utilisation levels whilst being interpretable and in line with operational intuition. Optimisation and uncertainty analysis are thus done using this model (Höppner, 2025).

5. Task 3 - Gradient Descent Cost Optimisation.

5.1 Fixed Demand Scenario

To determine the optimal levels of staffing, demand will be pegged at the historical average:

$\bar{D}$=96.22

The fitted cost model is then received as a functional of vans and employees alone.

5.2 Optimisation Method

The predicted cost function was minimised with gradient descent being done manually. The algorithm uses:

- Learning rate: 0.01
- 5,000 iterations
- Negativity imposed on vans and employees.

There were several initial values that were experimented around observed averages to provide convergence.

5.3 Optimal Staffing Levels

The continuous solution that results from optimisation is as follows:

Variable	Optimal Value
Vans	9.33
Employees	8.50
Predicted Cost (£)	1836.42

This implies that increased staffing seems to be the most ideal above the historical mean because the cost of penalty is very high in case of low capacity.

5.4 Integer Rounding

Practically, the level of staffing has to be an integer. When the vans and the employees are rounded to 9 vans and 9 employees respectively, a predicted cost of 1837.02 is obtained. The gap between the continuous optimum is very small showing that the solution is round-off insensitive.

6. Task 4 - Bootstrapping Uncertainty Quantification.

6.1 Bootstrap Procedure

To measure the uncertainty, the resampling of the daily records with replacement on 500 bootstrap samples was created (Dixon, 2006). For each bootstrap sample:

The last cost model was re-fitted.

The estimated cost within the optimal staffing levels was calculated.

This generated an empirical distribution of the expected costs.

6.2 Results

Statistic	Value (£)
Mean predicted cost	1837.06
2.5% quantile	1791.24
97.5% quantile	1887.22

6.3 Interpretation

The optimum staffing plan of 9 vans and 9 employees will give a cost of about 1837 per day with a confidence interval of 95 percent: 1791 to 1887.

The deviation is quite small, which means that the confidence in the prescribed staffing strategy is high.

7. Task 5: Reflection and Recommendations.

7.1 Key Findings

- Small overstaffing is significantly cheaper than understaffing.
- The leading factor in the penalty costs is utilisation.
- Real operational behaviour can only be captured by a nonlinear modelling.
- Stable staffing strategy is optimal in case of uncertainty.

7.2 Assumptions and Limitations

There were a number of simplifying assumptions. Linear regression presupposes constant error independence and variance. During optimisation, demand was assumed to be fixed and no distinction

was made on employees based on their role. The external interruptions like weather or traffic accidents were not explicitly modeled.

7.3 Suggested Improvements

The stochastic demand optimisation may be applied to future work, separate the concept of a driver and a loader, consider temporal factors including the variation by day of the week, and apply integer optimisation models. It would also be more realistic to extend the model to include multi-depot operations (Hasan et al., 2025).

8. Conclusion

The current study was able to formulate a statistically sound and operationally meaningful cost model of a delivery company. It combines regression modelling, gradient descent optimisation and bootstrap uncertainty analysis to give an insightful data-driven staffing recommendation. The findings are highly convincing in the proactive capacity planning as a viable strategy in reducing the expensive operational failures in the situation of uncertain demand.

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